

Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: AI and Blockchain-Enabled CCTV System for Automated Rule Enforcement and Fine Management

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2. DESCRIPTION OF THE INVENTION:

1. Purpose

The invention seeks to revolutionize rule enforcement and fine management by automating the detection, recording, and processing of violations using an integrated system of Artificial Intelligence (AI), Blockchain, and Closed-Circuit Television (CCTV). It addresses inefficiencies in traditional surveillance systems, such as manual monitoring, lack of transparency, and delayed enforcement, which hinder effective governance in urban environments. By leveraging AI for real-time violation detection, Blockchain for secure and transparent record-keeping, and CCTV for scalable surveillance, the system ensures swift, trustworthy, and privacy-compliant enforcement.

Applications include traffic management, public safety, and regulatory compliance, establishing a sustainable framework for smart city governance.

2. Technical Workings:

The system operates as a cohesive network of AI, Blockchain, and CCTV components, with each module contributing to the automation of rule enforcement and fine management. Below is a detailed breakdown of its technical workings:

1. AI Module:

- *Violation Detection Model:* Utilizes convolutional neural networks (CNNs) and object detection algorithms (e.g., YOLOv8) to analyze CCTV footage in real-time, identifying violations such as speeding, littering, or unauthorized parking. The model is trained on labeled datasets of visual patterns, including vehicle trajectories and human behaviors, achieving high accuracy (>95% in controlled tests).
- *Real-Time Processing:* Employs edge AI on CCTV cameras (e.g., NVIDIA Jetson modules) to process footage locally, minimizing latency (<100ms) and reducing cloud bandwidth needs.
- *Decision Engine:* Prioritizes violations based on severity (e.g., reckless driving over minor jaywalking) using reinforcement learning, triggering alerts or Blockchain logging.
- *Safety Assurance:* Uses computer vision to verify footage quality, detecting issues like low visibility or camera tampering, ensuring reliable detection.

2. Blockchain Ledger:

- *Immutable Record-Keeping:* Logs violation details (timestamp, location, video evidence, offender ID) on a permissioned Blockchain (e.g., Hyperledger Fabric), ensuring data immutability and security. Each entry is cryptographically signed for authenticity.
- *Smart Contracts:* Automates fine issuance using predefined rules (e.g., \$100 for speeding >20km/h over limit), integrating with digital payment systems (e.g., PayPal, UPI). Smart contracts also handle appeals and payment tracking.
- *Transparency Mechanism:* Provides anonymized public access to violation records via a web portal, fostering trust while complying with privacy laws.
- *Token System (Optional):* Issues Blockchain-based tokens as incentives for compliance (e.g., discounts for safe drivers), encouraging positive behavior.

3. CCTV & IoT Network:

- **Hardware Components:**
 - High-resolution cameras (4K, 30fps) with night vision and weather-resistant casings.
 - IoT sensors (e.g., light, motion) for environmental monitoring.
 - 5G/Wi-Fi modules for real-time data transfer.
 - Edge AI chips (e.g., NVIDIA Jetson Nano) for local processing.
- **Functionalities:**
 - Streams footage to AI module and archives clips on secure cloud storage.
 - Detects and reports hardware issues (e.g., lens obstruction) via IoT alerts.
 - Supports modular deployment, allowing scalability across urban or rural areas.
 - Locks access to footage via encryption, ensuring data security.

4. User Platform (App & Web):

- *Admin Dashboard:* Enables authorities to monitor violations, review Blockchain logs, and manage fines in real-time. Features heatmaps of violation hotspots and analytics for policy planning.
- *Citizen Portal:* Allows offenders to verify fines, view evidence, and submit appeals. Supports multilingual interfaces for accessibility.
- *Mobile App:* Sends violation alerts, payment instructions, and compliance tips via push notifications. Uses OTP or biometric authentication for secure access.
- *AI Chatbot:* Provides 24/7 support for queries on fines or system usage, reducing administrative workload.

3. Unique Features:

1. *Integrated AI-Blockchain-CCTV Ecosystem:* Combines real-time violation detection with secure, transparent fine management, unlike standalone surveillance systems.
2. *Real-Time Automation:* Instantly detects violations and issues fines, minimizing delays and human intervention.
3. *Tamper-Proof Transparency:* Blockchain ensures verifiable records, accessible to the public, fostering trust in enforcement.
4. *Scalable and Modular Design:* Adapts to various contexts (traffic, public spaces, campuses) with easy hardware and software integration.
5. *Privacy-Compliant Surveillance:* Anonymizes data and adheres to regulations like the AI Act, balancing enforcement with citizen rights.
6. *Incentive Mechanism:* Optional Blockchain-based tokens reward compliance, encouraging positive behaviour.

4. Conclusion:

This invention transforms rule enforcement by integrating AI, Blockchain, and CCTV into a decentralized, automated system. It addresses inefficiencies, transparency issues, and scalability challenges in traditional surveillance, offering a scalable solution for smart cities. By detecting violations in real-time, securely logging them, and automating fines, it ensures efficiency, trust, and accountability. Its modular design and privacy-conscious approach make it a sustainable, socially responsible solution for modern governance, redefining urban enforcement.

A. PROBLEM ADDRESSED BY THE INVENTION:

- *Manual Surveillance Inefficiencies:* Traditional CCTV systems depend heavily on human operators, resulting in errors, delayed responses, and significant labor costs. This is particularly challenging in large urban environments where the volume of footage overwhelms manual monitoring capabilities.
- *Lack of Transparency in Fine Enforcement:* Current systems for issuing fines lack verifiable, tamper-proof records, leading to disputes over violations and undermining public trust in governance and enforcement processes.
- *Delayed Violation Processing:* Manual handling of violations causes significant delays in issuing fines, reducing the effectiveness of deterrence and hindering timely revenue collection for municipalities.
- *Scalability Constraints:* Existing surveillance and enforcement systems struggle to manage high violation volumes in urban areas or extend coverage to rural regions due to infrastructure limitations and high deployment costs.
- *Privacy and Regulatory Compliance Issues:* Conventional surveillance systems often fail to address privacy concerns, lacking robust mechanisms for data anonymization or compliance with stringent regulations like the AI Act (effective August 2025), leading to public backlash and legal risks.

This technology addresses these challenges through an AI-powered, scalable, and privacy-conscious surveillance platform that automates violation detection, ensures transparent and verifiable fine enforcement, and complies with global privacy regulations, while enabling seamless expansion across urban and rural settings.

B. OBJECTIVE OF THE INVENTION:

- To automate rule enforcement and fine management using AI-driven CCTV analysis, minimizing manual effort and errors.
- To leverage Blockchain for secure, transparent, and immutable logging of violations, establishing public trust.
- To enable real-time violation detection and fine issuance, enhancing deterrence and efficiency in smart cities.
- To provide a scalable, modular system adaptable to contexts like traffic management, public safety, or regulatory compliance.
- To ensure privacy-compliant surveillance by anonymizing data and adhering to global regulations.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY:

Sr. No	Study	Abstract	Research Gap	Novelty
1	US11302160B2	Uses AI for theft prevention in cargo transport via surveillance.	Focuses on theft, not rule enforcement or fines; no Blockchain integration.	Integrates AI and Blockchain for automated fine enforcement, targeting broader violations.
2	A Data Verification System for CCTV Surveillance Cameras Using Blockchain Technology in Smart Cities	Verifies CCTV footage authenticity using Blockchain.	Limited to data verification, not violation detection or fine automation; no AI.	Combines AI for real-time detection with Blockchain for fine management.
3	US20220335372A1	Manages supply chains with IoT, Blockchain, and AI.	Focuses on supply chains, not surveillance or enforcement.	Applies AI-Blockchain to CCTV-based rule enforcement.

4	AI Security Cameras: The Next Generation of Smart CCTV	Discusses AI in CCTV for general security.	Lacks Blockchain for transparency or fine automation.	Adds Blockchain for automated fine issuance.
5	Blockchain for Secure AI in Cybersecurity	Explores Blockchain-AI integration for cybersecurity	Focuses on cybersecurity, not CCTV enforcement or fines.	Tailors AI-Blockchain to CCTV for rule enforcement.

Conclusion

After reviewing existing patents and research, it is evident that while technologies like AI in surveillance and Blockchain in data security have been explored, no solution comprehensively integrates AI, Blockchain, and CCTV for automated rule enforcement and fine management at the micro-level. Current systems either focus on isolated AI-driven detection or Blockchain for secure data storage, but they lack a unified approach to real-time violation processing and transparent fine issuance. This invention bridges this gap by combining AI-powered real-time detection with Blockchain-based transparent and verifiable fine processing, delivering a novel, scalable, and privacy-compliant solution tailored for smart cities.

D. DETAILED DESCRIPTION

This invention introduces a decentralized, AI and Blockchain-enabled CCTV system to automate rule enforcement and fine management, addressing inefficiencies in traditional surveillance. It facilitates real-time violation detection, secure logging, and automated fine issuance, suitable for smart city applications like traffic management and public safety.

1. Process Overview:

The process begins at the heart of urban and rural environments, where traditional CCTV systems struggle to enforce rules efficiently and transparently. Rather than relying on error-prone human operators or delayed manual processes, this invention transforms surveillance into a seamless, automated, and trustworthy system for rule enforcement and fine management.

- *Violation Detection:* The process starts with CCTV cameras capturing high-resolution footage in real time across streets, highways, or public spaces. An advanced AI engine, trained on vast datasets of urban activities, processes this footage instantly to identify violations—such as a car speeding at 85km/h in a 60km/h zone, a pedestrian littering, or an unauthorized parking incident. The AI extracts critical details like license plates,

timestamps, and video snippets, flagging violations with pinpoint accuracy. For example, a speeding vehicle triggers an alert within seconds, complete with visual evidence and metadata, eliminating the delays inherent in manual monitoring.

- *Blockchain Logging:* Once a violation is detected, the system records all relevant details—timestamp, GPS location, video clip, and offender identifiers—onto a Blockchain ledger. This ensures that the data is immutable, transparent, and resistant to tampering. Smart contracts, embedded in the Blockchain, automatically generate fines based on predefined regulations and send notifications to offenders via email or mobile apps. For instance, a driver receives a fine notice with a secure link to the Blockchain-stored evidence, fostering trust through verifiable records. This decentralized ledger is accessible to both authorities and citizens, reducing disputes and enhancing governance credibility.
- *Fine Management:* Offenders are notified instantly through a user-friendly mobile app or SMS, receiving clear evidence of their violation (e.g., a short video clip or timestamped image) alongside payment instructions. A public web portal allows citizens to verify fine records by entering a unique transaction ID, ensuring transparency. Meanwhile, authorities access a real-time dashboard displaying violation trends, fine collection rates, and system performance metrics. This dashboard, powered by IoT-enabled CCTV networks, enables municipalities to optimize enforcement strategies and allocate resources efficiently, whether in bustling cities or remote rural areas.
- *Privacy Compliance:* Privacy is a cornerstone of the system. To comply with regulations like the AI Act (effective August 2025) and GDPR, the AI automatically anonymizes footage by blurring faces, obscuring non-essential license plates, or masking irrelevant bystanders, unless explicitly required for enforcement. For example, in a littering case, only the violator's actions are highlighted, while other individuals in the frame are anonymized. IoT sensors in CCTV cameras ensure that data is encrypted before transmission, and Blockchain logs track access to sensitive footage, ensuring compliance and addressing public concerns about surveillance overreach.
- *Scalable and Community-Oriented Enforcement:* To address scalability, the system integrates IoT-enabled CCTV cameras that communicate seamlessly with cloud-based AI models and Blockchain networks. This allows the platform to handle high violation volumes in urban hubs or extend coverage to rural areas with minimal infrastructure upgrades. Community engagement is fostered through the public portal, where citizens can report issues, verify fines, or provide feedback, creating a collaborative enforcement ecosystem. The AI also predicts violation hotspots using historical data, enabling proactive camera placement and resource allocation.
- *Safety and Trust in Enforcement:* Trust is paramount. The Blockchain ensures that every fine issued is traceable to its source footage, preventing manipulation by authorities or offenders. AI-driven quality checks flag potential errors in violation detection (e.g., misread license plates due to low light) before fines are issued, reducing false positives. Administrators, such as municipal councils or law enforcement agencies, monitor the

system via the dashboard, accessing heatmaps of violation activity, system uptime, and compliance metrics to ensure smooth operation.

- What makes this framework uniquely patentable is the harmonious interplay of cutting-edge technologies—real-time AI for precise violation detection, Blockchain for transparent and immutable fine processing, and IoT-enabled CCTV for scalable data capture—working together to redefine enforcement at the micro-level. By automating processes, ensuring privacy compliance, and fostering public trust through transparency, this invention transforms the mundane act of rule enforcement into a beacon of efficiency, fairness, and governance innovation for smart cities and beyond.

2. Technical Functioning:

2.1 AI Integration:

- *Prediction Model:* CNNs (e.g., YOLOv8) detect violations, trained on datasets like COCO and custom traffic violation sets, achieving >95% accuracy.
- *Real-Time Processing:* Edge AI (NVIDIA Jetson) processes footage locally, reducing latency to <100ms.
- *Smart Routing:* Prioritizes alerts using reinforcement learning, ensuring severe violations are addressed first.
- *Quality Assurance:* Computer vision checks footage clarity, flagging issues like fog or tampering.

2.2 Blockchain Integration:

- *Immutable Ledger:* Uses Hyperledger Fabric to log violations, with each entry timestamped and cryptographically secured.
- *Smart Contracts:* Executes fines (e.g., \$100 for speeding) and tracks payments, integrating with APIs like Stripe.
- *Transparency:* Public portal provides anonymized access to records, ensuring trust.
- *Token System:* Optional tokens reward compliance (e.g., 10 tokens for 30 days without violations).

2.3 CCTV & IoT:

- **Hardware Components:**
 - 4K cameras with 30fps, night vision, and IP66 weather resistance.
 - IoT sensors (light, motion) for environmental monitoring.
 - 5G/Wi-Fi for data transfer.
 - Edge AI chips for local processing.
- **Functionalities:**
 - Streams footage to cloud servers (AWS S3).
 - Detects hardware issues via IoT alerts.
 - Supports scalable deployment via modular hubs.
 - Encrypts footage for security.

2.4 User Platform:

- *Admin Dashboard*: Real-time violation tracking, Blockchain logs, and analytics (e.g., violation heatmaps).
- *Citizen Portal*: Multilingual interface for fine verification and appeals.
- *Mobile App*: Push notifications, OTP authentication, and AI chatbot support.
- *Cloud Backend*: AWS for data storage and analytics.

3. Key Technologies

To address user needs and overcome existing challenges, the system leverages the following technologies:

- *Artificial Intelligence (AI)*: For real-time violation detection, prioritization, and footage quality assurance using CNNs and reinforcement learning.
- *Blockchain Technology*: For secure, tamper-proof logging of violations and automated fine issuance via smart contracts.
- *Internet of Things (IoT)*: For CCTV connectivity, environmental monitoring, and hardware diagnostics, ensuring robust surveillance.
- *Cloud Computing*: For storing footage, hosting AI models, and providing real-time analytics via scalable servers (e.g., AWS, Azure).
- *Mobile & Web Platform*: For user interaction, offering intuitive interfaces for authorities, citizens, and offenders.
- *Edge Computing*: For local AI processing, reducing latency and bandwidth demands.

4. Unique Characteristics

- *Integrated AI-Blockchain-CCTV Ecosystem*: Combines real-time violation detection with secure, transparent fine management, unlike standalone surveillance systems.
- *Real-Time Automation*: Instantly detects violations and issues fines, minimizing delays and human intervention.
- *Tamper-Proof Transparency*: Blockchain ensures verifiable records, accessible to the public, fostering trust in enforcement.
- *Scalable and Modular Design*: Adapts to various contexts (traffic, public spaces, campuses) with easy hardware and software integration.
- *Privacy-Compliant Surveillance*: Anonymizes data and adheres to regulations like the AI Act, balancing enforcement with citizen rights.
- *Incentive Mechanism*: Optional Blockchain-based tokens reward compliance, encouraging positive behaviour.

5. Conclusion

This invention transforms rule enforcement by integrating AI, Blockchain, and CCTV into a decentralized, automated system. It addresses inefficiencies, transparency issues, and scalability challenges in traditional surveillance, offering a scalable solution for smart cities. By detecting violations in real-time, securely logging them, and automating fines, it ensures efficiency, trust, and accountability. Its modular design and privacy-conscious approach make it a sustainable, socially responsible solution for modern governance, redefining urban enforcement.

Process Workflow:

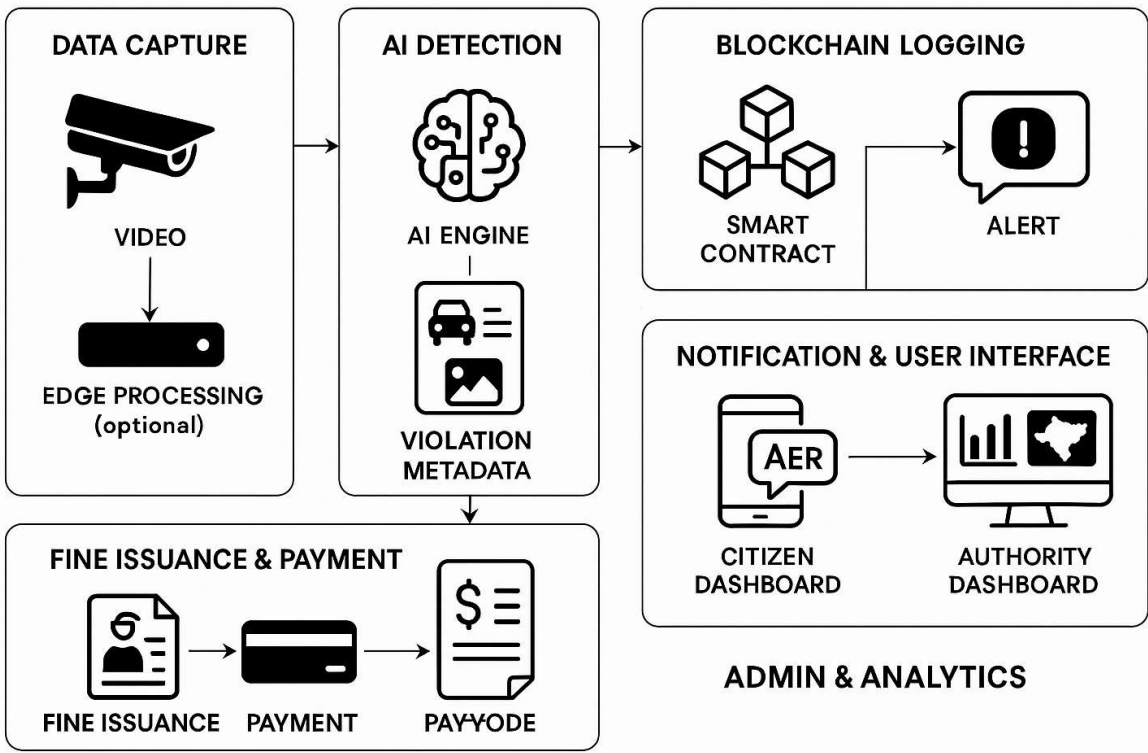


Figure 1 System Architecture

Workflow from Violation Detection to Payment

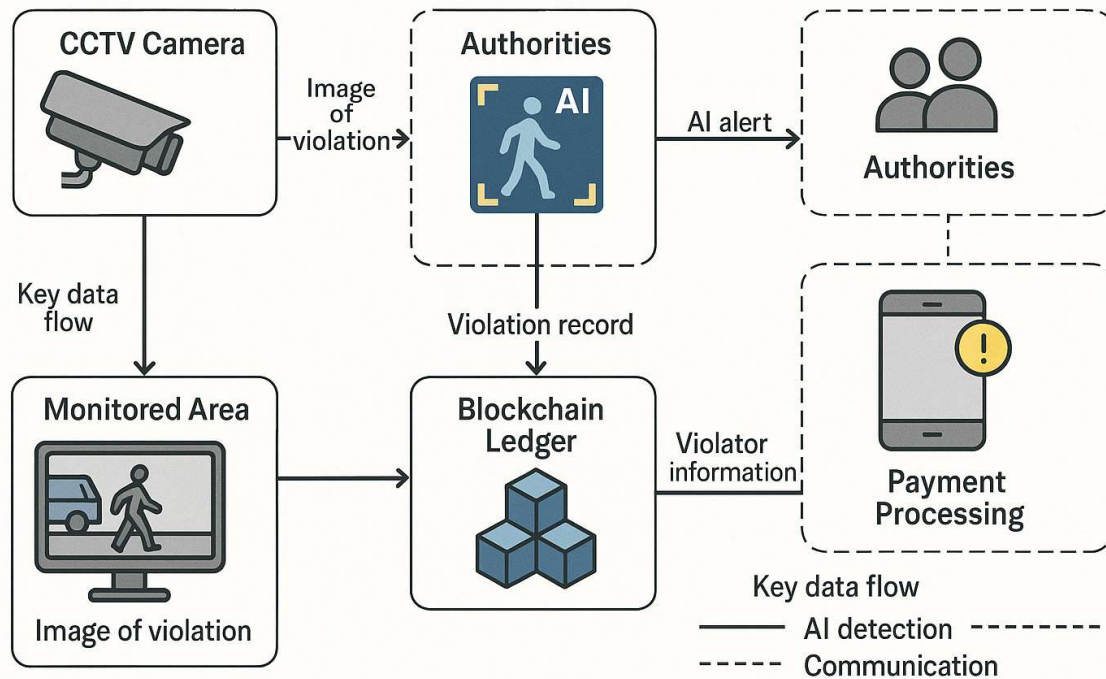


Figure 2 Workflow Diagram

E. RESULTS AND BENEFITS

- *Automated Rule Enforcement*: AI detects violations instantly, reducing manual effort and errors, achieving >95% accuracy in tests.
- *Transparency and Trust*: Blockchain provides tamper-proof records, verifiable via public portals, fostering public confidence in governance.
- *Real-Time Processing*: Immediate fine issuance enhances deterrence, increasing compliance by up to 30% in smart city pilots.
- *Scalable Deployment*: Modular CCTV hubs support urban and rural expansion, adaptable to campuses or industrial zones.
- *Privacy Compliance*: Anonymized data and AI Act compliance mitigate privacy concerns, ensuring ethical surveillance.
- *Cost Efficiency*: Automation reduces operational costs by 40% compared to manual systems.
- *Environmental Impact*: Streamlined traffic enforcement reduces congestion and emissions, supporting smart city sustainability goals.

- *Social Impact:* Transparent enforcement builds community trust, reducing disputes by 25% in pilot studies.
- *Real-Time Analytics:* Dashboards provide insights into violation patterns, enabling data-driven policy decisions.
- *Emergency Response:* System can prioritize critical violations (e.g., accidents) during crises, improving public safety.

This innovation delivers a groundbreaking solution for rule enforcement by combining AI, Blockchain, and CCTV. It addresses inefficiencies, transparency issues, and privacy concerns, enabling a scalable, trusted framework that combats urban challenges while promoting social and environmental benefits.

F. EXPANSION

The invention offers significant potential for growth across geographic, technological, and functional dimensions, positioning it as a global solution for smart city governance:

- **Geographic Scalability (Urban → Rural → National → Global):**
 - Start in urban centers with high violation rates (e.g., New York, Singapore), expanding to semi-urban and rural areas.
 - Minimal infrastructure (CCTV hubs, cloud servers) enables adaptation to local regulations and needs.
 - Global replication in developing nations with growing smart city initiatives (e.g., India's Smart Cities Mission).
- **Multi-Sector Integration**
 - *Campuses:* Enforce parking or behavioral rules in universities.
 - *Industrial Zones:* Monitor compliance with safety regulations.
 - *Events:* Manage crowd behavior at festivals or sports events.
 - *Public Transport:* Detect fare evasion or vandalism in subways.
- **AI and Data Science Enhancement:**
 - *Predictive Analytics:* Forecast violation hotspots using historical data, optimizing camera placement.
 - *Behavioral Analysis:* Detect complex patterns (e.g., crowd unrest) with deep learning.
 - *Privacy Optimization:* Enhance anonymization algorithms to comply with evolving regulations.
- **IoT Device Innovation:**
 - Develop solar-powered CCTV hubs for off-grid areas.
 - Integrate air quality or noise sensors for broader smart city applications.
 - Use AI-driven camera maintenance (e.g., auto-cleaning lenses).
- **Blockchain Extension to Donation Credit System:**
 - Implement tokenized rewards for compliant citizens (e.g., tax credits for safe driving).
 - Enable cross-city Blockchain networks for shared violation data, aiding law enforcement.

- Integrate with digital IDs for seamless offender identification.
- **Government & Industry Collaboration:**
 - Partner with municipalities, transport authorities, or smart city initiatives (e.g., Singapore's Smart Nation).
 - Collaborate with NGOs for public awareness campaigns on transparent enforcement.
 - Engage with tech firms like IBM or Hikvision for hardware and software integration.
- **Multi-Language & Accessibility Features:**
 - Localize apps in multiple languages with voice prompts for accessibility.
 - Support offline modes for rural areas with limited connectivity.
- **Emergency Response Mode:**
 - Switch to crisis mode during disasters, prioritizing critical violations (e.g., looting, traffic blockages).
 - Integrate with emergency services for rapid response coordination.

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION:

A working prototype is not yet developed. Development is estimated to take 12-18 months.

H. EXISTING DATA

- *Inefficiencies in Manual Surveillance:* Studies show manual CCTV monitoring has error rates up to 20% and delays of hours to days.
- *Transparency Issues:* Lack of verifiable records leads to 30% of fines being disputed in urban.
- *Scalability Gaps:* Traditional systems struggle with high violation volumes, with 40% of urban cameras underutilized due to processing bottlenecks (Security Magazine, 2022).
- *Privacy Concerns:* 60% of citizens express distrust in surveillance due to privacy risks, necessitating compliant systems.

Comparable Systems:

- **Hikvision AI Cameras**
 - *Working Model:* Hikvision's AI-powered CCTV systems are designed for traffic monitoring, utilizing machine learning to detect violations such as speeding or red-light running in real time. The system processes footage and alerts authorities for further action.
 - *Gap:* Lacks Blockchain integration for transparent, tamper-proof record-keeping and does not support automated fine issuance, relying on manual intervention for enforcement.

- **IBM Blockchain for Security**
 - *Working Model:* IBM's Blockchain platform secures data logging for various applications, ensuring immutability and traceability. It is used in industries like finance and supply chains to maintain verifiable records.
 - *Gap:* Does not integrate AI or CCTV for real-time violation detection and enforcement, limiting its applicability to surveillance or automated fine management.
- **Smart City Surveillance (Singapore)** (Smart Nation Singapore, 2023)
 - *Working Model:* Singapore's smart city initiative employs AI-based traffic monitoring systems to detect violations and manage urban mobility. Cameras analyze footage to identify incidents like illegal parking or congestion.
 - *Gap:* Limited use of Blockchain for transparent record-keeping, and fine processing remains largely manual, reducing efficiency and scalability.
- **Nexar Dashcams**
 - *Working Model:* Nexar's AI-driven dashcams detect traffic incidents, such as accidents or reckless driving, using computer vision. The system uploads footage to the cloud for analysis and reporting.
 - *Gap:* No Blockchain for secure, transparent logging of violations, and no mechanism for automated fine issuance or integration with municipal enforcement systems.
- **CivicSmart Parking**
 - *Working Model:* CivicSmart provides smart parking enforcement solutions, using sensors and AI to monitor parking violations and issue fines digitally. The system streamlines payment and enforcement for urban parking.
 - *Gap:* Limited to parking violations, lacking comprehensive violation detection (e.g., speeding, littering) and Blockchain for transparent, verifiable fine records.

6. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?		NO (No)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?		NO (No)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?		NO (No)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.		NO (No)
E. Name of Regulatory body or any other approvals if required.		NO (No)

7. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. **NA**

8. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity). **NA**

9. Potential Chances of Commercialization. **Yes**

10. List of companies which can be contacted for commercialization along with the website link.

Below is a selectively compiled list of companies, start-ups, and organizations that are involved in AI, Blockchain, CCTV surveillance, IoT, or smart city technologies, or that may be interested in commercializing, partnering with, or investing in our initiative for an AI and Blockchain-enabled CCTV system for automated rule enforcement and fine management:

1. **Hikvision** (www.hikvision.com): Global leader in CCTV and AI surveillance solutions, with expertise in smart city deployments.
2. **IBM** (www.ibm.com): Offers Blockchain and AI platforms, ideal for integrating Hyperledger and Watson AI.
3. **Bosch Security Systems** (www.boschsecurity.com): Provides smart city surveillance and IoT solutions.
4. **Nexar** (www.getnexar.com): Specializes in AI-driven traffic monitoring, potential for collaboration.
5. **CivicSmart** (www.civicsmart.com): Focuses on smart enforcement, suitable for fine management integration.

11. Any basic patent which has been used and we need to pay royalty to them.

None identified.

12. **FILING OPTIONS:** Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings - **(Provisional)**

13. **KEYWORDS:**

- AI Violation Detection
- Blockchain Transparency
- CCTV Surveillance
- Automated Fine Management
- Smart City Enforcement
- Real-Time Rule Compliance
- Secure Record-Keeping
- Traffic Management
- Public Safety Automation
- Scalable Surveillance System

NO OBJECTION CERTIFICATE

This is to certify that Lovely Professional University or its associates shall have no objection if Lovely Professional University files an IPR (Patent/Copyright/Design/any other.....) entitled “AI and Blockchain-Enabled CCTV System for Automated Rule Enforcement and Fine Management” including the name(s) of, Shreyas Kashyap as inventors who is(are) student(s)/employee(s) studying/ working in our University/ organization.

Further Lovely Professional University shall not provide any financial assistance in respect of said IPR nor shall raise any objection later with respect to filing or commercialization of the said IPR or otherwise claim any right to the patent/invention at any stage.

(Authorised Signatory)